Session 1 — Review of Fixed-Effects Regression

Assignment Questions and Solutions

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Solutions prepared for the course

The five assignment questions from "Session 1: Review of Fixed-Effects Regression" together with answers.

Throughout, we use the notation of the lecture: $\boldsymbol{y} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$, $\boldsymbol{C} = \boldsymbol{X}^\top \boldsymbol{X}$, $\mathrm{E}(\boldsymbol{\epsilon}) = \boldsymbol{0}$, $\mathrm{E}(\boldsymbol{\epsilon}\boldsymbol{\epsilon}^\top) = \sigma^2 \boldsymbol{I}_n$, and the spectral decomposition $\boldsymbol{C} = \boldsymbol{\Gamma}\boldsymbol{\Lambda}\boldsymbol{\Gamma}^\top$ with $\boldsymbol{\Gamma}$ orthogonal ($\boldsymbol{\Gamma}^\top\boldsymbol{\Gamma} = \boldsymbol{\Gamma}\boldsymbol{\Gamma}^\top = \boldsymbol{I}_p$) and $\boldsymbol{\Lambda} = \mathrm{diag}(\lambda_1, \dots, \lambda_p)$.

Q1. Centering eliminates the intercept

Question. Show that centering the observations $(\boldsymbol{x}, y)$ eliminates the intercept parameter from the linear regression model.

Solution

Consider the model with an explicit intercept,

$$y_i = \beta_0 + \boldsymbol{x}_i^\top \boldsymbol{\beta} + \epsilon_i, \qquad i = 1, \dots, n, \tag{1}$$

where $\boldsymbol{x}_i \in \mathbb{R}^p$ and $\boldsymbol{\beta} = (\beta_1, \dots, \beta_p)^\top$. Write this in matrix form using the augmented design matrix $\boldsymbol{X}_f = [\boldsymbol{1}_n, \boldsymbol{X}]$ and $\boldsymbol{\beta}_f = (\beta_0, \boldsymbol{\beta}^\top)^\top$, so $\boldsymbol{y} = \boldsymbol{X}_f \boldsymbol{\beta}_f + \boldsymbol{\epsilon}$.

Step 1 (normal equation for the intercept). The least-squares normal equations are $\boldsymbol{X}_f^\top \boldsymbol{X}_f \hat{\boldsymbol{\beta}}_f = \boldsymbol{X}_f^\top \boldsymbol{y}$. The first row of this system (corresponding to the column of ones) reads

$$\boldsymbol{1}_n^\top (\boldsymbol{y} - \boldsymbol{1}_n \hat{\beta}_0 - \boldsymbol{X}\hat{\boldsymbol{\beta}}) = 0 \implies n\hat{\beta}_0 = n\bar{y} - n\bar{\boldsymbol{x}}^\top \hat{\boldsymbol{\beta}}, \tag{2}$$

i.e.

$$\hat{\beta}_0 = \bar{y} - \bar{\boldsymbol{x}}^\top \hat{\boldsymbol{\beta}}, \qquad \bar{y} = \tfrac{1}{n}\sum_i y_i, \ \ \bar{\boldsymbol{x}} = \tfrac{1}{n}\sum_i \boldsymbol{x}_i. \tag{3}$$

So, whatever $\boldsymbol{\beta}$ is, the optimal intercept is fixed by (3): the intercept only ever adjusts for the sample means and carries no separate information about the effect of $\boldsymbol{x}$ on y.

Step 2 (profile out the intercept). Substitute (3) into the residual for observation i:

$$y_i - \hat{\beta}_0 - \boldsymbol{x}_i^\top \boldsymbol{\beta} = y_i - \bar{y} + \bar{\boldsymbol{x}}^\top \boldsymbol{\beta} - \boldsymbol{x}_i^\top \boldsymbol{\beta} = (y_i - \bar{y}) - (\boldsymbol{x}_i - \bar{\boldsymbol{x}})^\top \boldsymbol{\beta}. \tag{4}$$

Hence the least-squares objective, minimized over β_0 for fixed $\boldsymbol{\beta}$, becomes exactly the objective of a *no-intercept* regression of the centered response on the centered predictors:

$$S(\beta_0, \boldsymbol{\beta})\Big|_{\beta_0=\hat{\beta}_0(\boldsymbol{\beta})} = \sum_{i=1}^{n} \left[(y_i - \bar{y}) - (\boldsymbol{x}_i - \bar{\boldsymbol{x}})^\top \boldsymbol{\beta}\right]^2 = (\tilde{\boldsymbol{y}} - \tilde{\boldsymbol{X}}\boldsymbol{\beta})^\top (\tilde{\boldsymbol{y}} - \tilde{\boldsymbol{X}}\boldsymbol{\beta}), \tag{5}$$

where $\tilde{y}_i = y_i - \bar{y}$ and $\tilde{\boldsymbol{x}}_i = \boldsymbol{x}_i - \bar{\boldsymbol{x}}$ are the centered observations ($\tilde{\boldsymbol{X}}$ has columns summing to zero).

Step 3 (conclusion). Minimizing this centered sum of squares over $\boldsymbol{\beta}$ gives

$$\hat{\boldsymbol{\beta}} = (\tilde{\boldsymbol{X}}^\top \tilde{\boldsymbol{X}})^{-1} \tilde{\boldsymbol{X}}^\top \tilde{\boldsymbol{y}}, \tag{6}$$

which is identical to the slope estimate obtained from fitting the model without an intercept to the centered data $(\tilde{\boldsymbol{x}}_i, \tilde{y}_i)$. The intercept plays no role once the data are centered. It is recovered afterward, if needed, from $\hat{\beta}_0 = \bar{y} - \bar{\boldsymbol{x}}^\top \hat{\boldsymbol{\beta}}$, but it does not enter or affect the estimation of $\boldsymbol{\beta}$. This is exactly the convention used throughout Session 1, where the model is written as $\boldsymbol{y} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$ with the understanding that the data have already been centered.

Numerical check (R)

```r
## Q1: numerical check -- centering eliminates the intercept
set.seed(1)
n <- 30; p <- 3
X <- matrix(rnorm(n * p), n, p)
beta_true <- c(1.5, -2, 0.7)
beta0_true <- 4
y <- beta0_true + X %*% beta_true + rnorm(n, sd = 0.5)

## (a) fit with intercept, using X as is
fit_full <- lm(y ~ X)

## (b) center y and X, fit WITHOUT intercept
yc <- y - mean(y)
Xc <- scale(X, center = TRUE, scale = FALSE)
fit_centered <- lm(yc ~ Xc - 1)

cat("Slope estimates, model WITH intercept (raw X):\n")
print(round(coef(fit_full)[-1], 4))

cat("\nSlope estimates, model WITHOUT intercept (centered X, y):\n")
print(round(coef(fit_centered), 4))

cat("\nRecovered intercept beta0_hat = ybar - xbar'beta_hat :",
    round(mean(y) - as.numeric(colMeans(X) %*% coef(fit_centered)), 4), "\n")
cat("Direct intercept estimate from full model          :",
    round(coef(fit_full)[1], 4), "\n")
```

Output:

```
Slope estimates, model WITH intercept (raw X):
     X1      X2      X3
 1.5036 -2.0847 0.7323
```

```
Slope estimates, model WITHOUT intercept (centered X, y):
    Xc1     Xc2     Xc3
 1.5036 -2.0847 0.7323

Recovered intercept beta0_hat = ybar - xbar'beta_hat : 4.0641
Direct intercept estimate from full model          : 4.0641
```

The slope estimates from the two fits agree exactly, and the intercept recovered from (3) matches the one estimated directly, confirming the derivation.

Q2. Tikhonov regularization

Question. What is Tikhonov regularization? Why do you put constraint?

Solution

Tikhonov regularization (Tikhonov, 1943; Tikhonov and Arsenin, 1977) is a general method for solving ill-posed inverse (least-squares) problems by adding a penalty on the size of the solution. For the linear model $\boldsymbol{y} = \boldsymbol{X\beta} + \boldsymbol{\epsilon}$, instead of minimizing the unconstrained sum of squares $S(\boldsymbol{\beta}) = \|\boldsymbol{y} - \boldsymbol{X\beta}\|^2$, one minimizes the *penalized* criterion

$$S_\Gamma(\boldsymbol{\beta}) = \|\boldsymbol{y} - \boldsymbol{X\beta}\|^2 + \|\boldsymbol{\Gamma}_T\boldsymbol{\beta}\|^2, \tag{7}$$

for a chosen Tikhonov matrix $\boldsymbol{\Gamma}_T$ (not to be confused with the eigenvector matrix $\boldsymbol{\Gamma}$ used elsewhere; here $\boldsymbol{\Gamma}_T$ is a design choice, often $\boldsymbol{\Gamma}_T = \sqrt{k}\,\boldsymbol{I}_p$). The solution is

$$\hat{\boldsymbol{\beta}}_T = (\boldsymbol{X}^\top\boldsymbol{X} + \boldsymbol{\Gamma}_T^\top\boldsymbol{\Gamma}_T)^{-1}\boldsymbol{X}^\top\boldsymbol{y}. \tag{8}$$

When $\boldsymbol{\Gamma}_T = \sqrt{k}\boldsymbol{I}_p$ this reduces exactly to the ridge estimator introduced in the lecture,

$$\hat{\boldsymbol{\beta}}(k) = (\boldsymbol{C} + k\boldsymbol{I}_p)^{-1}\boldsymbol{X}^\top\boldsymbol{y}, \qquad \boldsymbol{C} = \boldsymbol{X}^\top\boldsymbol{X}, \tag{9}$$

so ridge regression is the special (isotropic) case of Tikhonov regularization. Equivalently, this is the solution to the constrained optimization problem

$$\min_{\boldsymbol{\beta}} \ \|\boldsymbol{y} - \boldsymbol{X\beta}\|^2 \quad \text{subject to} \quad \|\boldsymbol{\Gamma}_T\boldsymbol{\beta}\|^2 \leq t, \tag{10}$$

for some $t > 0$ in one-to-one correspondence with k (Lagrangian duality).

Why impose the constraint? There are several closely related reasons, all touched on in the lecture:

(i) **Non-invertibility / ill-conditioning of $\boldsymbol{C}$.** If $\operatorname{rank}(\boldsymbol{X}) < p$ (in particular whenever $p \geq n$, the high-dimensional case), $\boldsymbol{C} = \boldsymbol{X}^\top\boldsymbol{X}$ is singular and $\hat{\boldsymbol{\beta}} = \boldsymbol{C}^{-1}\boldsymbol{X}^\top\boldsymbol{y}$ does not exist. Even when $\boldsymbol{C}$ is technically invertible but has a very large condition number $\kappa(\boldsymbol{C}) = \lambda_{\max}/\lambda_{\min}$ (multicollinearity), the OLS solution is numerically unstable. Adding $k\boldsymbol{I}_p$ shifts every eigenvalue by k, so $\boldsymbol{C} + k\boldsymbol{I}_p$ is always invertible for $k > 0$ and well-conditioned for k not too small.

(ii) **Variance control (bias–variance trade-off).** As shown in the lecture, $\operatorname{tr}\operatorname{Var}(\hat{\boldsymbol{\beta}}) = \sigma^2 \sum_j \lambda_j^{-1}$ blows up when eigenvalues are small. The penalty shrinks the estimator ($\hat{\boldsymbol{\beta}}(k) \to \mathbf{0}$ as $k \to \infty$), introducing bias but sharply reducing variance; for suitably chosen k this yields a strictly smaller mean-squared error than the unbiased OLS estimator (see Q4 and the MSE trade-off frame in the lecture).

(iii) **Regularizing an ill-posed inverse problem.** More generally (this is the classical Tikhonov motivation from inverse problems), a small perturbation in $\boldsymbol{y}$ can cause an enormous change in $\hat{\boldsymbol{\beta}}$ when $\boldsymbol{C}$ is nearly singular; the penalty term makes the map $\boldsymbol{y} \mapsto \hat{\boldsymbol{\beta}}(k)$ continuous and stable (well-posed in the sense of Hadamard).

(iv) **Encoding prior information.** A nonzero $\boldsymbol{\Gamma}_T$ can encode structural beliefs (e.g., smoothness, a preferred norm, or a Bayesian Gaussian prior on $\boldsymbol{\beta}$ with precision matrix $\boldsymbol{\Gamma}_T^\top \boldsymbol{\Gamma}_T / \sigma^2$); the isotropic ridge case corresponds to an i.i.d. Gaussian prior on each coefficient.

Q3. Verifying $\operatorname{Var}(\hat{\boldsymbol{\theta}}) = \sigma^2 \boldsymbol{\Lambda}^{-1}$

Question. Verify the equality $\operatorname{Var}(\hat{\boldsymbol{\theta}}) = \sigma^2 \boldsymbol{\Lambda}^{-1}$ (Eq. (35) in the lecture).

Solution

Recall the canonical regression model obtained by rotating the design matrix with the orthogonal matrix $\boldsymbol{\Gamma}$ from the spectral decomposition $\boldsymbol{C} = \boldsymbol{\Gamma}\boldsymbol{\Lambda}\boldsymbol{\Gamma}^\top$:

$$\boldsymbol{y} = \boldsymbol{Z}\boldsymbol{\theta} + \boldsymbol{\epsilon}, \qquad \boldsymbol{Z} = \boldsymbol{X}\boldsymbol{\Gamma}, \quad \boldsymbol{\theta} = \boldsymbol{\Gamma}^\top \boldsymbol{\beta}. \tag{11}$$

Step 1: $\boldsymbol{Z}^\top \boldsymbol{Z} = \boldsymbol{\Lambda}$. Using $\boldsymbol{\Gamma}^\top \boldsymbol{\Gamma} = \boldsymbol{I}_p$,

$$\boldsymbol{Z}^\top \boldsymbol{Z} = \boldsymbol{\Gamma}^\top \boldsymbol{X}^\top \boldsymbol{X} \boldsymbol{\Gamma} = \boldsymbol{\Gamma}^\top \boldsymbol{C} \boldsymbol{\Gamma} = \boldsymbol{\Gamma}^\top (\boldsymbol{\Gamma}\boldsymbol{\Lambda}\boldsymbol{\Gamma}^\top) \boldsymbol{\Gamma} = (\boldsymbol{\Gamma}^\top \boldsymbol{\Gamma}) \boldsymbol{\Lambda} (\boldsymbol{\Gamma}^\top \boldsymbol{\Gamma}) = \boldsymbol{\Lambda}. \tag{12}$$

Step 2: OLS estimator of $\boldsymbol{\theta}$. Since $\boldsymbol{Z}^\top \boldsymbol{Z} = \boldsymbol{\Lambda}$ is invertible (diagonal, with $\lambda_j > 0$ assumed), the OLS estimator is

$$\hat{\boldsymbol{\theta}} = (\boldsymbol{Z}^\top \boldsymbol{Z})^{-1} \boldsymbol{Z}^\top \boldsymbol{y} = \boldsymbol{\Lambda}^{-1} \boldsymbol{Z}^\top \boldsymbol{y}. \tag{13}$$

Step 3: variance. Because $\boldsymbol{\Lambda}^{-1}$ and $\boldsymbol{Z}$ are non-random (fixed design) and $\operatorname{Var}(\boldsymbol{y}) = \operatorname{Var}(\boldsymbol{\epsilon}) = \sigma^2 \boldsymbol{I}_n$,

$$\begin{aligned}
\operatorname{Var}(\hat{\boldsymbol{\theta}}) &= \boldsymbol{\Lambda}^{-1} \boldsymbol{Z}^\top \operatorname{Var}(\boldsymbol{y})\, \boldsymbol{Z} \boldsymbol{\Lambda}^{-1} = \sigma^2 \boldsymbol{\Lambda}^{-1} \boldsymbol{Z}^\top \boldsymbol{Z} \boldsymbol{\Lambda}^{-1} && (14)\\
&= \sigma^2 \boldsymbol{\Lambda}^{-1} \boldsymbol{\Lambda} \boldsymbol{\Lambda}^{-1} \qquad \text{(using Step 1)} && (15)\\
&= \sigma^2 \boldsymbol{\Lambda}^{-1}. \qquad \blacksquare && (16)
\end{aligned}$$

Because $\boldsymbol{\Lambda}$ is diagonal, this simply says $\operatorname{Var}(\hat{\theta}_j) = \sigma^2 / \lambda_j$ and $\operatorname{Cov}(\hat{\theta}_j, \hat{\theta}_{j'}) = 0$ for $j \neq j'$ — the rotated (canonical) coefficients are uncorrelated, each with variance inversely proportional to its eigenvalue, which is exactly why small λ_j's are dangerous.

Numerical check (R)

```r
set.seed(7)
n <- 200; p <- 5
X <- matrix(rnorm(n * p), n, p)
sigma2 <- 2

C <- t(X) %*% X
eig <- eigen(C, symmetric = TRUE)
Gamma <- eig$vectors
Lambda <- eig$values

Z <- X %*% Gamma
ZtZ <- t(Z) %*% Z
cat("Q3: Z'Z (should equal diag(Lambda)) :\n")
print(round(ZtZ, 4))

Var_theta_formula <- sigma2 * diag(1 / Lambda)
Var_theta_direct <- sigma2 * solve(ZtZ)

cat("\nmax abs difference between formula sigma^2*Lambda^{-1} and direct sigma^2*(Z'Z)
    ^{-1}:\n")
print(max(abs(Var_theta_formula - Var_theta_direct)))
```

Output:

```
Q3: Z'Z (should equal diag(Lambda)) :
         [,1]     [,2]    [,3]     [,4]     [,5]
[1,] 230.5069   0.0000   0.000   0.0000   0.0000
[2,]   0.0000 226.7861   0.000   0.0000   0.0000
[3,]   0.0000   0.0000 193.575   0.0000   0.0000
[4,]   0.0000   0.0000   0.000 165.2526   0.0000
[5,]   0.0000   0.0000   0.000   0.0000 147.8797

max abs difference between formula sigma^2*Lambda^{-1} and direct sigma^2*(Z'Z)^{-1}:
[1] 2.255141e-17
```

$\boldsymbol{Z}^\top \boldsymbol{Z}$ is numerically diagonal with entries equal to the eigenvalues of $\boldsymbol{C}$, and $\sigma^2 \boldsymbol{\Lambda}^{-1}$ agrees with the directly computed $\sigma^2(\boldsymbol{Z}^\top \boldsymbol{Z})^{-1}$ up to floating-point error ($\sim 10^{-17}$).

Q4. Proving $\operatorname{tr} \operatorname{Var}(\hat{\boldsymbol{\beta}}(k)) = \sigma^2 \sum_j \lambda_j/(\lambda_j + k)^2$

Question. Prove Equation (36) in the lecture.

Solution

Recall the ridge estimator $\hat{\boldsymbol{\beta}}(k) = \boldsymbol{C}(k)^{-1} \boldsymbol{X}^\top \boldsymbol{y}$ with $\boldsymbol{C}(k) = \boldsymbol{C} + k\boldsymbol{I}_p$, and $\operatorname{Var}(\boldsymbol{y}) = \sigma^2 \boldsymbol{I}_n$.

Step 1: covariance matrix of $\hat{\boldsymbol{\beta}}(k)$. Since $\boldsymbol{C}(k)^{-1}\boldsymbol{X}^\top$ is a fixed matrix,

$$\begin{aligned}
\operatorname{Var}(\hat{\boldsymbol{\beta}}(k)) &= \boldsymbol{C}(k)^{-1}\boldsymbol{X}^\top \operatorname{Var}(\boldsymbol{y})\, \boldsymbol{X}\boldsymbol{C}(k)^{-1} = \sigma^2\, \boldsymbol{C}(k)^{-1}\boldsymbol{X}^\top \boldsymbol{X}\boldsymbol{C}(k)^{-1} && (17)\\
&= \sigma^2\, \boldsymbol{C}(k)^{-1}\boldsymbol{C}\,\boldsymbol{C}(k)^{-1}, \qquad (\text{using } \boldsymbol{X}^\top\boldsymbol{X} = \boldsymbol{C}). && (18)
\end{aligned}$$

Step 2: spectral form of $\boldsymbol{C}(k)^{-1}$. From $\boldsymbol{C} = \boldsymbol{\Gamma}\boldsymbol{\Lambda}\boldsymbol{\Gamma}^\top$,

$$\boldsymbol{C}(k) = \boldsymbol{C} + k\boldsymbol{I}_p = \boldsymbol{\Gamma}\boldsymbol{\Lambda}\boldsymbol{\Gamma}^\top + k\boldsymbol{\Gamma}\boldsymbol{\Gamma}^\top = \boldsymbol{\Gamma}(\boldsymbol{\Lambda} + k\boldsymbol{I}_p)\boldsymbol{\Gamma}^\top, \tag{19}$$

so, since $\boldsymbol{\Lambda} + k\boldsymbol{I}_p$ is diagonal and invertible for $k > 0$,

$$\boldsymbol{C}(k)^{-1} = \boldsymbol{\Gamma}(\boldsymbol{\Lambda} + k\boldsymbol{I}_p)^{-1}\boldsymbol{\Gamma}^\top. \tag{20}$$

Step 3: substitute into (18). Using $\boldsymbol{\Gamma}^\top\boldsymbol{\Gamma} = \boldsymbol{I}_p$ repeatedly,

$$\begin{aligned} \boldsymbol{C}(k)^{-1}\boldsymbol{C}\,\boldsymbol{C}(k)^{-1} &= \left[\boldsymbol{\Gamma}(\boldsymbol{\Lambda} + k\boldsymbol{I}_p)^{-1}\boldsymbol{\Gamma}^\top\right]\left[\boldsymbol{\Gamma}\boldsymbol{\Lambda}\boldsymbol{\Gamma}^\top\right]\left[\boldsymbol{\Gamma}(\boldsymbol{\Lambda} + k\boldsymbol{I}_p)^{-1}\boldsymbol{\Gamma}^\top\right] && (21) \\ &= \boldsymbol{\Gamma}(\boldsymbol{\Lambda} + k\boldsymbol{I}_p)^{-1}\boldsymbol{\Lambda}(\boldsymbol{\Lambda} + k\boldsymbol{I}_p)^{-1}\boldsymbol{\Gamma}^\top. && (22) \end{aligned}$$

Because $\boldsymbol{\Lambda}$, $(\boldsymbol{\Lambda}+k\boldsymbol{I}_p)^{-1}$ are diagonal (hence commute), the middle diagonal matrix has j-th diagonal entry $\lambda_j/(\lambda_j + k)^2$. Write $\boldsymbol{D}_k = \operatorname{diag}\big(\lambda_j/(\lambda_j + k)^2\big)_{j=1}^p$, so

$$\operatorname{Var}(\hat{\boldsymbol{\beta}}(k)) = \sigma^2\,\boldsymbol{\Gamma}\boldsymbol{D}_k\boldsymbol{\Gamma}^\top. \tag{23}$$

Step 4: trace. Using the cyclic property of the trace and $\boldsymbol{\Gamma}^\top\boldsymbol{\Gamma} = \boldsymbol{I}_p$,

$$\operatorname{tr}\operatorname{Var}(\hat{\boldsymbol{\beta}}(k)) = \sigma^2\operatorname{tr}(\boldsymbol{\Gamma}\boldsymbol{D}_k\boldsymbol{\Gamma}^\top) = \sigma^2\operatorname{tr}(\boldsymbol{D}_k\boldsymbol{\Gamma}^\top\boldsymbol{\Gamma}) = \sigma^2\operatorname{tr}(\boldsymbol{D}_k) = \sigma^2\sum_{j=1}^p \frac{\lambda_j}{(\lambda_j + k)^2}. \quad \blacksquare \tag{24}$$

As a sanity check, setting $k = 0$ recovers $\operatorname{tr}\operatorname{Var}(\hat{\boldsymbol{\beta}}) = \sigma^2\sum_j \lambda_j^{-1}$, matching the OLS result used to motivate the ridge trick in the lecture.

Numerical check (R)

```r
k <- 3.5
Ck_inv <- solve(C + k * diag(p))
Var_beta_k_direct <- sigma2 * Ck_inv %*% C %*% Ck_inv
trace_direct <- sum(diag(Var_beta_k_direct))

trace_formula <- sigma2 * sum(Lambda / (Lambda + k)^2)

cat("\nQ4: trace via direct matrix formula :", round(trace_direct, 6), "\n")
cat("Q4: trace via closed-form sum_j lambda_j/(lambda_j+k)^2 :", round(trace_formula,
    6), "\n")
```

Output:

```
Q4: trace via direct matrix formula : 0.051452
Q4: trace via closed-form sum_j lambda_j/(lambda_j+k)^2 : 0.051452
```

The direct matrix computation $\operatorname{tr}\left[\sigma^2\boldsymbol{C}(k)^{-1}\boldsymbol{C}\boldsymbol{C}(k)^{-1}\right]$ and the closed-form sum agree exactly, confirming the proof.

Q5. High-dimensional case ($p > n$): R code and five predictions

Question. Write the R code for a high-dimensional case $p > n$ and make five predictions.

Solution / discussion

When $p > n$, $\boldsymbol{C} = \boldsymbol{X}^\top \boldsymbol{X}$ has rank at most $n < p$, so it is necessarily singular and the ordinary least squares estimator $\hat{\boldsymbol{\beta}} = \boldsymbol{C}^{-1}\boldsymbol{X}^\top \boldsymbol{y}$ does not exist (Remedy R1/R2 of the lecture). We use **ridge regression** (Remedy R2, $\alpha = 0$ in `glmnet`), which replaces $\boldsymbol{C}$ by $\boldsymbol{C}(k) = \boldsymbol{C} + k\boldsymbol{I}_p$, always invertible for $k > 0$, exactly as derived in Q2–Q4. The ridge penalty k (glmnet calls it λ) is chosen by 10-fold cross-validation. We then simulate 5 new (out-of-sample) covariate vectors and report ridge predictions for them.

```r
## Q5: High-dimensional case p > n, ridge regression, five predictions
set.seed(2026)
suppressMessages(library(glmnet))

n <- 40     # sample size
p <- 100    # number of predictors, p > n
s0 <- 8     # number of truly active predictors

X <- matrix(rnorm(n * p), n, p)
beta_true <- rep(0, p)
beta_true[1:s0] <- rnorm(s0, mean = 2, sd = 0.5)
sigma <- 3
eps <- rnorm(n, sd = sigma)
y <- as.numeric(X %*% beta_true + eps)

cat("n =", n, " p =", p, " (p > n, so X^T X is singular)\n")
cat("rank(X^T X) <=", min(n, p), " while p =", p, "\n\n")

## 10-fold CV to choose ridge parameter (alpha = 0 => ridge)
cvfit <- cv.glmnet(X, y, alpha = 0, nfolds = 10)
cat("Selected ridge lambda (lambda.min):", round(cvfit$lambda.min, 4), "\n")
cat("Selected ridge lambda (lambda.1se):", round(cvfit$lambda.1se, 4), "\n\n")

fit <- glmnet(X, y, alpha = 0, lambda = cvfit$lambda.min)

## 5 new observations for prediction
Xnew <- matrix(rnorm(5 * p), 5, p)
ynew_true_mean <- as.numeric(Xnew %*% beta_true)
preds <- predict(fit, newx = Xnew)

results <- data.frame(
  obs = 1:5,
  true_mean = round(ynew_true_mean, 3),
  predicted = round(as.numeric(preds), 3)
)
print(results, row.names = FALSE)

cat("\nNumber of nonzero coefficients kept by ridge (all, by construction):",
    sum(coef(fit) != 0) - 1, "out of", p, "\n")
```

Output:

```
n = 40 p = 100 (p > n, so X^T X is singular)
rank(X^T X) <= 40 while p = 100

Selected ridge lambda (lambda.min): 43.7872
```

```
Selected ridge lambda (lambda.1se): 244.8057

 obs true_mean predicted
   1     2.703    -0.358
   2    -2.980     0.454
   3     2.466    -1.184
   4    -9.377    -2.432
   5     1.643    -0.185

Number of nonzero coefficients kept by ridge (all, by construction): 100 out of 100
```

Remarks

- By construction $n = 40 < p = 100$, so $\boldsymbol{C}$ is exactly rank-deficient ($\text{rank}(\boldsymbol{C}) \leq 40$); OLS via `lm` would report `NA` coefficients for the $p - n$ redundant directions. Ridge, by contrast, always returns a unique, well-defined estimate because $\boldsymbol{C}(k)$ is full rank for any $k > 0$.
- Ridge does not perform variable selection. As printed above, all $p = 100$ coefficients remain nonzero (unlike the lasso, $\alpha = 1$), which is exactly the eigenvalue-shrinkage picture of Q4 — every direction is shrunk, none is set exactly to zero.
- The five predictions are noticeably shrunk toward zero relative to the true conditional means $\boldsymbol{x}_{\text{new}}^\top \boldsymbol{\beta}_{\text{true}}$; this is the expected bias–variance trade-off from Q2/Q4: with only $n = 40$ observations and $\sigma = 3$ noise driving $p = 100$ parameters, a fairly large $\lambda \approx 44$ was selected by cross-validation, which reduces variance substantially at the cost of noticeable shrinkage bias in individual predictions.